Karthik Thyagarajan

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Education

Purdue University West Lafayette, Indiana
B.S. of Computer Science & Artificial Intelligence (Double Major), GPA: 3.93 *May 2027*

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Programming in C, Competitive Programming, OOP

Experience

AI Research Intern May 2026 – Aug 2026
Samsung Research America *Mountain View, CA*

- Built an on-device signal router for always-on ambient AI on Bixby Edge AI, running inference on the phone to filter incoming device signals at 95% suppression and 90% intent-routing accuracy so expensive frontier models are only called for events that need them; owned the router, task design, and evaluation harness.
- Co-architected Samsung's always-on AI agent orchestrator from scratch in Kotlin, owning half of a 25+ module runtime that ships on XR glasses and gives phone, glasses, and third-party developers a single platform to build AI agents on.
- Built the agent safety and guardrail layer that governs what an agent is allowed to do, including permission checks on every tool call, user approval flows for risky actions, spending limits per session, and tamper-evident audit logs with automatic PII redaction.
- Built the model routing layer that matches each request to the right model, from small on-device models up to frontier APIs, with streaming responses, mid-request cancellation, and automatic failover between providers, plus the agent's long-term memory and conversation history systems.
- Shipped computer-use style phone control letting agents read the screen and operate any app directly through taps, typing, and navigation, with user confirmation before risky actions, plus native Android integrations for calendar, alarms, location, and health data.

Machine Learning Engineering Intern Jun 2025 – May 2026
Peraton Labs *Silver Spring, MD*

- Developed a first-of-its-kind reinforcement learning agent for IoT malware detection in PyTorch, granting it low-level network traversal actions and full state observability to search device environments a brute-force scan cannot cover, cutting exploration latency 35% and raising detection coverage 25% across 500K+ daily device events.
- Engineered the agent's feature set from binary signatures, file paths, filenames, and static analysis output, improving malware classification over single-signal baselines, delivered inside a classified defense research environment.
- Built large-scale ETL pipelines and a heterogeneous graph architecture with graph neural networks and autoencoders to model device-to-device communication patterns, accelerating policy convergence 40%.

Computer Vision Researcher Feb 2025 – May 2025
Memories.ai *Remote*

- Built a video memory framework for hours-long video that vision language models cannot process directly, combining keyframe extraction, dynamic segmentation, CLIP indexing, and YOLO and SAM object and face detection to index 10K+ streams in Python, Flask, and PostgreSQL, with 60% throughput improvement from frame sampling optimization.
- Implemented multimodal search over the index, combining transcript, visual, and entity embeddings with query rewriting, RAG-style semantic retrieval, and graph traversal across scenes, objects, and named people.
- Published pymavi, the company's Python SDK, to PyPI with 2K+ downloads, implementing async processing, retry and backoff handling, and REST API integration for video analysis workflows.

Undergraduate Robotics Researcher Mar 2025 – May 2025
IDEAS Lab *West Lafayette, IN*

- Built a real-time SLAM pipeline in C++ and Python with RGB-D sensor fusion and Kalman filtering, streaming calibrated camera data to a multi-GPU server, improving 3D reconstruction accuracy 25% and reducing latency 30%.

- Implemented neural radiance fields (NeRF) for novel view synthesis, generating photorealistic 3D scene reconstructions from camera data for autonomous navigation.
- Generated real-time navigation signals from incomplete 3D maps, choosing where a robot should explore next in environments it has not fully mapped.

Data Engineering Intern Aug 2024 – Dec 2024
AgRPA *West Lafayette, IN*

- Built an end-to-end weed detection pipeline processing 200GB+ of drone imagery in Python with parallel ETL workflows and partitioned cloud storage, cutting analysis time 40%.
- Fine-tuned YOLO segmentation models to 92% accuracy on 50K+ images by fusing RGB and infrared camera streams, reducing herbicide application 60% and saving $150K annually.
- Designed the aerial data collection protocol, tuning flight altitude, speed, and camera configuration against motion blur and ground resolution limits, and oversampling frame rate to build a reusable training corpus.

Projects & Clubs

Repple | *Python (FastAPI), PostgreSQL, Supabase, Pydantic, OpenAI API, Anthropic Claude, TypeScript* Ongoing

- Co-founder of a competitive fitness iOS app live on the App Store with 200+ active users; built the entire backend in FastAPI, PostgreSQL, and Supabase with Pydantic-validated schemas.
- Built real-time synchronized multiplayer workout sessions, a custom ELO matchmaking algorithm pairing users by fitness level, and cron-driven point aggregation powering weekly team matchups, streaks, and leaderboards.
- Shipped LLM-generated workout plans using OpenAI and Anthropic Claude that adapt to each user‘s logged performance metrics.

google-tools-mcp | *TypeScript, Node.js, MCP SDK, Google Workspace APIs, OAuth 2.0, npm* Ongoing

- Built and published an MIT-licensed MCP server to npm in TypeScript and Node.js, unifying 169 tools across Drive, Docs, Sheets, Slides, Gmail, Calendar, and Forms behind a single OAuth 2.0 flow with auto re-auth and dynamic scope handling.
- Designed read-before-edit guards, lazy authentication, and multi-account profile support so agents can safely operate across an entire Google Workspace from one context, with one-command setup and no telemetry.

Veritas (Catapult 2026 - Best Proof-of-Human Application) | *Next.js 14, React 19, PostgreSQL, Supabase, NextAuth.js,*

- Built a clinical research platform pairing World ID iris-biometric proof-of-personhood with GPT-4o and Pinecone response-quality scoring to filter fraudulent trial participants, targeting the 4% of volunteers who fabricate symptoms for compensation across a $35B market.
- Designed the three-layer identity, scoring, and researcher-review pipeline in Next.js 14 and PostgreSQL; won Best Proof-of-Human Application at Catapult 2026.

Caladrius (HackGT - 2nd Place) | *React Native, Python, LangGraph, AWS S3, REST API, Next.js* Sep 2025

- Architected a privacy-first AI triage assistant with encrypted, zero-knowledge data transfer for HIPAA compliance, built in React Native, Python, and Next.js on AWS S3 storage.
- Built a multi-agent diagnostic system with LangGraph reaching 85% triage accuracy on symptom analysis, designed with input from practicing physicians and medical filing professionals; 2nd place in HackGT 12‘s social impact track.

Technical Skills

Languages: Python, Kotlin, TypeScript, Java, C/C++, JavaScript, SQL
Frameworks & Platforms: FastAPI, Next.js, React, React Native, Node.js, Flask, Android SDK, PostgreSQL, Supabase
Cloud & DevOps: AWS (S3, EC2, Lambda), GCP, Vercel, Docker, Git, CI/CD, Linux, REST API, OAuth 2.0
AI/ML: LLMs (RAG, agents, MCP, LangChain, LangGraph, RLHF), Agent orchestration & tool-calling, On-device / edge inference, PyTorch, Reinforcement Learning, Graph Neural Networks, Computer Vision (YOLO, SAM, CLIP, pose estimation), NeRF & SLAM, Diffusion, GAN